

Transformări - liniare

Fie U, V' K -sp vectoriale, K corp comutativ

$f: U \rightarrow V'$ s.m. transform liniară dacă

$$\begin{cases} f(x+y) = f(x) + f(y) \\ f(\alpha x) = \alpha f(x) \end{cases}$$

$\forall x, y \in U, \forall \alpha \in K$

$$\Leftrightarrow f(\alpha x + \beta y) = \alpha f(x) + \beta f(y) \quad \forall x, y \in U, \forall \alpha, \beta \in K$$

Ex: Fie U, V_1, V_2 K -sp vect, $f: U \rightarrow V_1, g: U \rightarrow V_2$

functia $h: U \rightarrow V_1 \times V_2, h(x) = (f(x), g(x)) \quad \forall x \in U$

Aristotele că h este transform liniară $\Leftrightarrow f, g$ transform lin

sol: h transform lin $\Leftrightarrow \forall \alpha, \beta \in K \quad \forall x, y \in U$

$$h(\alpha x + \beta y) = \alpha h(x) + \beta h(y)$$

$$\Leftrightarrow \forall \alpha, \beta \in K \quad \forall x, y \in U$$

$$(f(\alpha x + \beta y), g(\alpha x + \beta y)) =$$

$$= \alpha (f(x), g(x)) + \beta (f(y), g(y))$$

$$\Leftrightarrow \text{---} : f(\alpha x + \beta y), g(\alpha x + \beta y) =$$

$$= (\alpha f(x), \alpha g(x)) + (\beta f(y), \beta g(y))$$

$$\Leftrightarrow (f(\alpha x + \beta y), g(\alpha x + \beta y)) = (\alpha f(x) + \beta f(y), \alpha g(x) + \beta g(y))$$

$$\Leftrightarrow \text{---} : \begin{cases} f(\alpha x + \beta y) = \alpha f(x) + \beta f(y) \\ g(\alpha x + \beta y) = \alpha g(x) + \beta g(y) \end{cases}$$

$$\Leftrightarrow f, g \text{ transform lin}$$

Generalizare: Fie $V, V_1, \dots, V_m$ K -sp. vect., $m \in \mathbb{N}^*$

Stăm că: $V_1 \times V_2 \times \dots \times V_m = \{(x_1, \dots, x_m) : x_i \in V_i, \dots, x_m \in V_m\}$
este K -sp. vect.

Fie $f_i: V \rightarrow V_i$ $i = \overline{1, m}$ funcții

$h: V \rightarrow V_1 \times \dots \times V_m$, $h(x) = (f_1(x), \dots, f_m(x)) \forall x \in V$

$h: V \rightarrow V_1 \times \dots \times V_m$ este K -lin. dacă și numai dacă $f_1, \dots, f_m$ sunt K -lin.

ex: a) Fie $m \in \mathbb{N}^*$ și $f: \mathbb{R}^m \rightarrow \mathbb{R}$. Arătați că f este transf. lin. dacă și numai dacă există $a_1, \dots, a_m \in \mathbb{R}$ unice def.

$$f(x_1, \dots, x_m) = a_1 x_1 + a_2 x_2 + \dots + a_m x_m, \forall (x_1, \dots, x_m) \in \mathbb{R}^m$$

b) Determinați transf. lin. $g: \mathbb{R}^m \rightarrow \mathbb{R}^n$, $n, m \in \mathbb{N}^*$

Sol: a) " $\Rightarrow$ " Existenta $a_1, \dots, a_m$

Fie $(x_1, \dots, x_m) \in \mathbb{R}^m$

$$(x_1, \dots, x_m) = (x_1, 0, \dots, 0) + (0, x_2, \dots, 0) + \dots + (0, \dots, 0, x_m) =$$

$$= x_1 \underset{e_1}{(1, 0, \dots, 0)} + x_2 \underset{e_2}{(0, 1, \dots, 0)} + \dots + x_m \underset{e_m}{(0, \dots, 0, 1)}$$

$$(x_1, \dots, x_m) = x_1 e_1 + \dots + x_m e_m, \text{ unde } e_i = \underset{i}{(0, \dots, 0, 1, 0, \dots, 0)} \quad i = \overline{1, m}$$

$$f(x_1, \dots, x_m) \stackrel{\text{f.lin.}}{=} \underbrace{x_1 f(e_1)}_{a_1} + \dots + \underbrace{x_m f(e_m)}_{a_m}$$

Unicitate: P. $\exists b_1, \dots, b_m \in \mathbb{R}$

$$\text{c.} \quad f(x_1, \dots, x_m) = b_1 x_1 + \dots + b_m x_m$$

$$a_i = f(e_i) = b_i \cdot 0 + \dots + b_{i-1} \cdot 0 + b_i \cdot 1 + \dots \cdot 0 = b_i, i = \overline{1, m}$$

$$\Leftarrow \text{ " } \forall \alpha, \beta \in \mathbb{R}, (x_1, \dots, x_m), (y_1, \dots, y_m) \in \mathbb{R}^m$$

$$\begin{aligned} f(\alpha(x_1, \dots, x_m) + \beta(y_1, \dots, y_m)) &= f(\alpha x_1 + \beta y_1, \dots, \alpha x_m + \beta y_m) \\ &= a_1(\alpha x_1 + \beta y_1) + \dots + a_m(\alpha x_m + \beta y_m) \\ &= \alpha(a_1 x_1 + \dots + a_m x_m) + \beta(a_1 y_1 + \dots + a_m y_m) \\ &= \alpha f(x_1, \dots, x_m) + \beta f(y_1, \dots, y_m) \Rightarrow f \text{ transf lin} \end{aligned}$$

$$b) \text{ } \forall m, n \in \mathbb{N}^*, f: \mathbb{R}^m \rightarrow \mathbb{R}^n$$

$$\mathbb{R}^m \xrightarrow{f} \mathbb{R}^n \xrightarrow{p_j} \mathbb{R}$$

projecti
canonice

$$\left\{ \begin{array}{l} p_j: \mathbb{R}^n \rightarrow \mathbb{R} \\ p_j(y_1, \dots, y_n) = y_j, j = \overline{1, n} \end{array} \right. \quad (p_j \circ f) = f_j$$

$$f(x_1, \dots, x_m) = (f_1(x_1, \dots, x_m), \dots, f_n(x_1, \dots, x_m))$$

$$f \text{ transf lin} \Leftrightarrow f_1, \dots, f_n \text{ transf lin} \Leftrightarrow$$

$$\exists! a_1^1, \dots, a_m^1, a_1^2, \dots, a_m^2, a_1^3, \dots, a_m^3, \dots, a_1^n, \dots, a_m^n \in \mathbb{R}$$

$$\text{or } f(x_1, \dots, x_m) = (a_1^1 x_1 + \dots + a_m^1 x_m, a_1^2 x_1 + \dots + a_m^2 x_m, \dots, a_1^n x_1 + \dots + a_m^n x_m) =$$

$$= \begin{bmatrix} a_1^1 & a_2^1 & \dots & a_m^1 \\ a_1^2 & a_2^2 & \dots & a_m^2 \\ \vdots & \vdots & \ddots & \vdots \\ a_1^n & a_2^n & \dots & a_m^n \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix}$$

prop transf lin $\in$
prop matricial

ex = ~~Arbitr~~ & exists a transf R lin

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}^2 \text{ a.t. } f(1,1) = (2,5), \text{ a. } f(1,0) = (1,4)$$

Esté f isomorfism?

Sol: $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ transf lin $\Leftrightarrow \exists a_1, a_2, a_3, a_4 \in \mathbb{R}$

$$\text{a.t. } f(x,y) = (a_1x + a_2y, a_3x + a_4y)$$

$$= \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$f(1,1) = (2,5) \Leftrightarrow (a_1 + a_2, a_3 + a_4) = (2,5)$$

$$f(1,0) = (1,4) \Leftrightarrow (a_1, a_3) = (1,4) \Rightarrow (a_2, a_4) = (1,1)$$

$$\Rightarrow f(x,y) = \begin{bmatrix} 1 & 1 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \quad \forall (x,y) \in \mathbb{R}^2$$

$$f \text{ bij} \Leftrightarrow \forall (x_2, y_2) \in \mathbb{R}^2 \exists! (x_1, y_1) \in \mathbb{R}^2 \text{ a.t. } f(x_1, y_1) = (x_2, y_2)$$

$$\Leftrightarrow \begin{bmatrix} 1 & 1 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}, \quad \begin{vmatrix} 1 & 1 \\ 4 & 1 \end{vmatrix} = -3 \neq 0$$

$$\Leftrightarrow \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 4 & 1 \end{bmatrix}^{-1} \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$$

ex = For $T: \mathbb{R}_2(\mathbb{R}) \rightarrow \mathbb{R}^2$

$$T(f) = (f(1), f(-1)) \quad \forall f \in \mathbb{R}_2(\mathbb{R})$$

$$\left(\begin{array}{l} E_x = \mathbb{R}[x] \rightarrow \mathbb{R}, \quad E_x(f) = f(x) \\ \Rightarrow T(f) = (E_1(f), E_{-1}(f)) \\ \text{transf lin} \end{array} \right) \Rightarrow T \text{ transf lin}$$

Ker T = ? Im T = ? Esté T inj, surj?

Defn: $f: V \rightarrow V'$ K-linear

$$f \text{ inj} \Leftrightarrow \ker f = \{0\}$$

" $\Rightarrow$ " Pp f linf. For $x \in \text{Ker } f \Leftrightarrow f(x) = 0 = f(0) \Rightarrow \boxed{x=0}$
 $\Leftrightarrow \text{Ker } f = \{0\}$

" $\Leftarrow$ " P_p Kerf $= \{0\}$. Für $x, y \in V$ o. $f(x) = f(y) \Leftrightarrow$

$$\Leftrightarrow f(x) - f(y) = 0 \Leftrightarrow f(x - y) = 0 \Leftrightarrow x - y \geq 0$$

trans of him

$$x = y$$

Ugenj

für $f \in \text{Ker } T$: $\begin{cases} f = a_2 x^2 + a_1 x + a_0, & a_0, a_1, a_2 \in \mathbb{R} \\ T(f) = 0 \end{cases}$

$$\tau(p) = 0 \quad (\equiv)$$

$$\Leftrightarrow (Q_2 + Q_1 + Q_0, Q_2 - Q_1 + Q_0) = (0, 0) \Leftrightarrow$$

$$\begin{aligned} & \begin{cases} a_2 + a_1 + a_2 = 0 \\ a_2 - a_1 + a_0 = 0 \end{cases} \\ & \hline & \text{[scribbles]} \end{aligned}$$

$$\Rightarrow S = \{(-a_2, 0, a_2) \mid a_2 \in \mathbb{R}\}$$

$$-a_1x^2 + a_2 = a_2 \in \mathbb{R} = \langle x^2 - 1 \rangle = \text{Ker } T$$

$\Rightarrow$ Truierung

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$$2a_1 = 0 \Rightarrow a_1 = 0$$

$$2Q_2 + 2Q_0 = 0$$

$$Q_2 \neq Q_0 = 0$$

$$\Rightarrow Q_0 = -Q_2$$

T surj $\Leftrightarrow \text{Im } T = \mathbb{R}^2$

$$\mathbb{R}^2 \subseteq \text{Term} \quad \text{Int} T = \{T(f) : f \in \underbrace{\mathbb{R}^2}_{\mathbb{R}^2} \} = \{ (Q_2 + Q_1 + Q_0, Q_2 - Q_1 + Q_0) : Q_0, Q_1, Q_2 \in \mathbb{R} \}$$

Dem

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$\forall (x,y) \in \mathbb{R}^2. \textcircled{?} a_0, a_1, a_2 \in \mathbb{R} \text{ s.t. } \begin{cases} a_2 + a_1 + a_0 = x \\ a_2 - a_1 + a_0 = y \end{cases}$ compatible

$\Rightarrow \mathbb{R}^2 \subseteq \text{Im } T \subseteq \mathbb{R}^2 \Rightarrow \text{Im } T = \mathbb{R}^2 \Rightarrow T \text{ surj}$

$f(0) = f(1) = 0$

$\Rightarrow (p) = (q) \text{ s.t. } 0 \vee 0 \text{ s.t. } x \neq 0 \text{ s.t. } \{0\} = \{0\} \text{ of } \Rightarrow$

$0 = p \Rightarrow 0 = (p-1) \Rightarrow 0 = (p-1) \Rightarrow$

$\textcircled{p=1}$ not found

$p=1$

$\exists a_0, a_1, a_2 \text{ s.t. } a_2 + a_1 + a_0 = 1 \text{ s.t. } T \text{ s.t. } 0 = (1)T$

$\left. \begin{aligned} 0 &= a_2 + a_1 + a_0 \\ 0 &= a_2 - a_1 + a_0 \end{aligned} \right\} \Rightarrow (0,0) = (a_2 + a_1 + a_0, a_2 - a_1 + a_0) \Rightarrow$

$\begin{aligned} 0 &= 1 \cdot 0 = 0 = 1 \cdot 0 \\ 0 &= a_2 + a_1 + a_0 \\ 0 &= a_2 - a_1 + a_0 \\ 0 &= a_2 + a_0 \end{aligned}$

$T(1) = \langle 1, 1 \rangle = (1, 1) \text{ s.t. } 1 \cdot 1 = 1$
 $\textcircled{\text{Im } T = \mathbb{R}^2}$

$\mathbb{R} = \text{Im } T \Rightarrow \text{Im } T = \mathbb{R}$

$\{(a_2 + a_1 + a_0, a_2 - a_1 + a_0)\} = \{(1, 1)\} \Rightarrow \{1, 1\} = \text{Im } T \Rightarrow \mathbb{R}$